Random Forest Classification

Classification example • Jupyter Notebook-style code and output

In [1]:

```
import matplotlib.pyplot as plt
from sklearn.datasets import make_classification
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, confusion_matrix
```

In [2]:

```
X, y = make_classification(n_samples=180, n_features=2,
                           n_redundant=0, n_informative=2,
                           n_clusters_per_class=1,
                           class_sep=1.35, random_state=7)
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, random_state=42, stratify=y
)
```

Out [ ]:

```
Training samples: 135
Test samples: 45
```

In [3]:

```
# Train a random forest classifier
model = RandomForestClassifier(n_estimators=100, random_state=42)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)

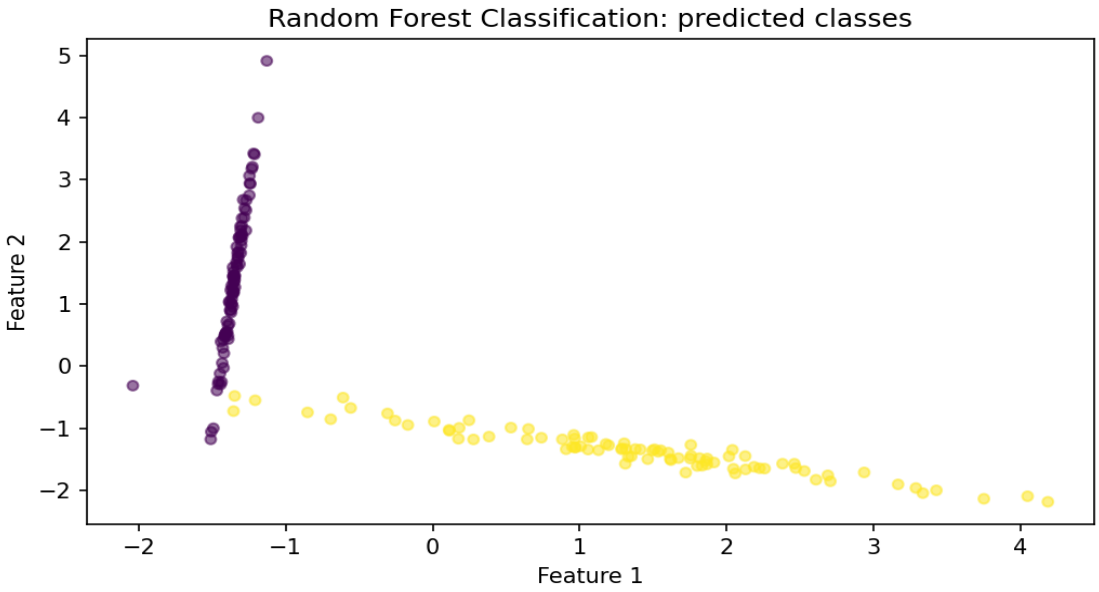
print("Accuracy:", round(accuracy_score(y_test, y_pred), 3))
print("Confusion matrix:\n", confusion_matrix(y_test, y_pred))
```

Out [ ]:

```
Accuracy: 0.978
Confusion matrix:
[[23  0]
 [ 1 21]]
```

In [4]:

```
# Visualize predicted classes
plt.scatter(X[:, 0], X[:, 1], c=model.predict(X), alpha=0.55, s=22)
plt.xlabel('Feature 1')
plt.ylabel('Feature 2')
plt.title('Random Forest Classification: Predicted Classes')
plt.show()
```



Rendered output of the plotting cell